

Giovani Diaz Alfaro

Cognitive Scientist

Orlando, FL – Open to Relocation

Creative and detail-oriented scientist with 8 years of experience in interdisciplinary research and development. Experienced in project management, workflow development, research study execution, data analysis, and publication with a strong focus on synthesizing knowledge from multiple disciplines to address complex, challenging issues and questions.

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EDUCATION

**MSc Interdisciplinary Studies
(conc. Cognitive Systems,
Ethics)**

University of Central Florida
TBA: Summer 2026

GradCert. Cognitive Sciences

University of Central Florida
Spring 2023

**BA Philosophy
(min. Cognitive Sciences)**

University of Central Florida
Spring 2020

SOFTWARE

Scripting and Statistics: Python,
R, MATLAB

Graphics and Diagrams: Adobe
Photoshop

Webdev: JavaScript, WordPress,
basic HTML5

Surveys and Studies: Qualtrics

Publishing: LaTeX

LANGUAGES

Bilingual: English • Spanish

Basic Proficiency: French

PROFESSIONAL EXPERIENCE

Research Associate

Cognitive Sciences Laboratory, University of Central Florida | Orlando (FL), USA
June 2023 – Present

United States Department of Defense, NASA, Minerva Research Initiative:
Understanding Cognition in Support of Future Space Missions
(PI: S.M. Fiore)

- Develop unobtrusive measures of team functioning for future space missions by leveraging multi-sensor data (e.g., physiological, environmental, etc.) to monitor individual and team cognitive states.
- Implemented suggestions to enhance operational environment workflow and design for long-duration space exploration missions with human and artificially intelligent (e.g. onboard systems) team members.

Lockheed Martin Corporation: Externalized and Extended Cognition: *Cognitive Offloading for Manned-Unmanned Teaming*
(PI: S.M. Fiore)

- Provide recommendations for mitigating workload in human-machine teaming contexts (e.g. AI as teammate), as well as offer a multi-disciplinarily informed set of theoretically grounded research questions and designs for understanding and improving workload in manned-unmanned teaming environments.

- Conduct interdisciplinary research on individuals and teams to evaluate cognition and develop interventions to reduce workload and improve performance.
- Interpret and understand the specific needs and problem areas for industries /companies such as the United States Department of Defense, Lockheed Martin, the United States Air Force, and more.
- Perform research focused on extended cognition, its theoretic foundation and principles, and practical applications in contexts such as virtual reality, space travel, military operations, and more.
- Apply to and analyze the role of ethical theories and principles in complex contemporary situations.
- Perform and apply human-machine systems research principles in interdisciplinary contexts.
- Manage lab teams and experiment logistics needs, human subjects for research, and maintenance of lab/experimental environments.
- Propose research for funding, design and conduct experiments, and analyze complex data with the end goal to publish results.
- Train in, and prepare for future work in teaching, research, project management, and consulting.

KEY SKILLS

Experimentation:

Human-Subjects Research, Design, Planning, Workflow Management, Surveying, Quality Assurance Testing

Collaboration:

Multi-disciplinary research, multi-disciplinary project management

Soft Skills: Public Speaking, Collaboration, Creative Thinking, Concept Presentation

Other: Team Member/Employee Management, Computer Hardware

CERTIFICATIONS

Python 3 (Codecademy, 2022)

Inclusive and Ethical Leadership Certificate (Leadership Summit & USF, 2024)

RESEARCH

INTERESTS

- Practical Cognitive Science
- Human Enhancement
- Team Cognition
- Interdisciplinarity
- Human-Machine Teaming
- Software for Scientific Teams
- Moral Cognition
- Ethics of Artificial Intelligence
- Critical Pedagogy

Supervisor, Process Improvement Coordinator

Orlando Health Imaging Centers | Orlando (FL), USA

Sep 2020 – Dec 2021

Project(s): Quality Assurance Program, St. Petersburg Expansion, EPIC Systems Launch and Training

- Demonstrated critical thinking skills; identified & fully documented 'key' clinical and non-clinical processes; defined associated key measures/metrics; documented performance and recommended improvements. Designed/led initiatives throughout organization for process improvement.
- Performed research and analyzed data against internal standards and external benchmarks to identify opportunities for improvement. Increased departmental productivity statistics by 40%, varying organizational process efficacies, and analyzed organizational error frequencies. Designed and implemented measures and execution processes that minimized errors by 60%.
- Exercised effective Leadership: assured (1) resources, (2) training and (3) leadership communication for team/team members and organizational success.
- Continuously researched trends in operational data to identify optimization, development, and efficiency opportunities.. Used Excel to create and send reports on various projects and productivity measures across departments. (E.g. Productivity reports, auditing and error tracking, revenue/collections tracking, potential revenue vs actual revenue, process efficacy analyses)
- Performed ongoing review of key processes; gathered/analyzed data, documented/developed process diagrams; presented findings, evaluated improvement options, eliciting buy-in from all stakeholders.

Bartender

Orlando (FL), USA

September 2018 – Present

- Anticipated/satisfied customer needs while providing an exceptional experience.
- Work efficiently and masterfully/consistently produce craft cocktails in a fast-paced setting.
- Exceed expectations in selling whiskeys, wine, seasonal specials, and total nightly sales.
- Led other bartenders to complete their duties and ensure smooth operations throughout the shift.
- Manage financials and payments to employees.
- Keep up with, and exceed expectations cleaning, organizing, opening/closing, and general maintenance of the bar throughout every shift.
- Back up all roles, including cook and any front of house position.

PROFESSIONAL DEVELOPMENT

Team Learning for Better Scientific Software (TL4BSSw)

Brought together experts across disciplines and domains to develop an open knowledge base with tools to support and evaluate team learning in the interdisciplinary groups who produce scientific software. The products of the workshop contribute to understanding of collaborative learning in scientific software development and use, and are open to all in service of better scientific software.

Related manuscript in preparation:

Newton, O. B., **Diaz Alfaro, G.**, Raybourn, E. M., Curfman McInnes, L., Lenhardt, C., Aydeniz, M., Fruckaj, H., Zhou, H., Milewicz, R., Serhan, Z., Almutairi, M., Ramchandra Patole, S. (in preparation). Team learning for better scientific software.

Target: IEEE Computing in Science and Engineering

EPIC System Launch (Orlando Health Imaging Centers)

Managed, trained, and led resource construction for the organization-wide launch and integration of EPIC's healthcare software system.

RELATED EXPERIENCE

Guest Lecturer

Topics in Cognitive Science (Fall '24/'25)

[UCF, Instructor: S. M. Fiore; ≤15 graduate students]

Research Methods in Cognitive Science (Spring '25)

[UCF, Instructor: S. M. Fiore; ≤15 graduate students]

Teaching Assistant

Ethics in Science & Technology (Spring '20)

(UCF, Instructor: D. Jones)

Mentorship

Sophia Sakakibara Capello, *B.Sc. Psychology, Minor: Cognitive Sciences* / Accepted to STEM Summer Scholars Institute at Indiana University Bloomington (IUB) & Indiana University-Purdue University Indianapolis (IUPUI)

Reviewer

Conference: Human Factors and Ergonomics Society International Annual Meeting (2024-26)

PUBLICATIONS

Conference Proceedings

Diaz Alfaro, G., Fiore, S. M., & Oden, K. (2024). Externalized and extended cognition: Cognitive Offloading for human-machine teaming. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*.

<https://doi.org/10.1177/10711813241275506>

Presentations, Panels, and Posters

Diaz Alfaro, G. (2024, November). Team Learning for a Better Foundation for Scientific Software. Position Paper presented at the 2024 Team Learning for Better Scientific Software Workshop, Orlando, FL.

Sonnenfeld, N., Nguyen, B., **Diaz Alfaro, G.**, Daniliv, V., Newton, O. B., Jentsch, F., Fiore, S. M. (2024). Eliciting Requirements and Recommendations for Training Macrocognition in Teams: Considerations for Collaborative Problem Solving. *Poster presented at 2024 ASPIRE—the HFES International Annual Meeting, Phoenix, AZ.*